Laura Akullian D'Agostino, Ph.D.

Vice President, Chemistry at Monimoi Therapeutics

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Laura Akullian D'Agostino, Ph.D., is Vice President of Chemistry at Monimoi Therapeutics. Over two decades at Mersana, Avila, Celgene, Bristol Myers Squibb, and Dunad, she has led chemistry for programs in oncology, immunology, neurodegeneration, and cell therapy, working across small molecules, covalent modulators, protein degraders, and antibody-drug conjugates.

Laura trained as a synthetic organic chemist, working on enantioselective catalysis with Amir Hoveyda for her Ph.D. at Boston College, after a B.S. in chemistry at State University of New York at Oswego with research in natural product isolation.

Employment

August 2025–Present

Monimoi Therapeutics
Vice President, Chemistry

- Series A close for startup focused on covalency, proteomics, and oncology, third employee building capabilities in synthetic, medicinal and computational chemistry

June 2023–August 2025

Dunad Therapeutics
Vice President, Chemistry

- Expanded covalent and targeted protein degradation platform
- Progressed first program from initiation to CNS-penetrant development candidate for ALS in 18 months
- Responsible for Medicinal Chemistry, Biochemistry, Proteomics, Computational Chemistry, Protein Sciences, and Compound Management in the US and UK, including outsourcing to China, India, and Canada
- Established R&D budget working with C-suite and consultants
- Presented to investors and pharma VCs to raise funds in a competitive market

2019–June 2023

Bristol Myers Squibb
Director, Small Molecule Drug Discovery

- Led team of 8 internal and 18 external chemists and chemical biologists
- Established next-generation ADC platform including licensing deal with Tubulis for cysteine-selective bioconjugation chemistry
- Supported early clinical development of first-in-class dual DGKα/ζ inhibitor (Phase 1, NCT05407675)
- Implemented small molecule strategies to improve durability and transfection efficiency for engineered T cell therapies
- Led academic collaborations with Professors David MacMillan and Robert Knowles at Princeton University

2012–2019

Celgene
Principal Scientist → Director

- Built small molecule portfolio in immuno-oncology working remotely with immunologists and strategic leaders
- Built and implemented state-of-the-art platform in heterobifunctional degradation

- Served on Joint Scientific Steering Committees for Nimbus (Tyk2 inhibitor NDI-034858) and Ikena (AHR antagonist IK-175)
- Managed internal chemists in Cambridge and San Diego and external chemistry in China

2009–2012

Avila Therapeutics
Senior Scientist → Principal Scientist

- Delivered covalent CNS-penetrant isoform-selective HER2 inhibitor for HER2+ CNS-metastatic breast cancer
- Delivered covalent isoform-selective FGFR4 inhibitor for hepatocellular carcinoma
- Led programs within a multi-target collaboration with Sanofi Oncology

2006–2009

Mersana Therapeutics
Scientist I → Scientist II

- Led first fully internal program from inception to IND: anti-angiogenic fumagillin-derived polymer conjugate XMT-1107
- Designed novel small molecule payloads, variable release rate linkers, and bioconjugates

Education

2001–2006

Boston College
Ph.D. in Organic Chemistry, Advisor: ***Amir H. Hoveyda***

- Dissertation: Discovery and Development of New Catalysts for the Enantioselective Synthesis of α-Chiral Amines and Tertiary β-Ketoalcohols
- Donald J. White Teaching Excellence Award

1997–2001

State University of New York at Oswego
B.S. Cum Laude, Advisor: Joseph LeFevre

- Research: Natural product isolation from NY state plant life
- Gamma Sigma Epsilon Chemistry Honor Society

2018

Duke University Management School
21-week Project Leader Effectiveness Program

Awards, membership & service

Keystone: Covalency in Chemical Probe and Drug Discovery

- 2027 (planned): Organizer and session chair
- 2026: Proposal submission

Cancer Prevention & Research Institute of Texas

- 2023–present: Grant reviewer seed, discovery & development stage

Med Chem Boston

- 2027: Networking event organizer and host

Gordon Research Seminar

- 2022–present: Medicinal Chemistry GRS Faculty

Covalent Drug Discovery Summit

- 2023, 2025: Chair and workshop leader

American Association for Cancer Research

- 2019–2023: Chemistry in Cancer Research (CICR) Steering Committee
- 2018–Present: CICR Working Group

American Chemical Society

- 2022 ACS Mid-Career Investigator Award
- 2000–Present: Member, Division of Organic Chemistry

Diversity, Equity & Inclusion

- Wellesley Girls in STEM
- Women in the Enterprise of Science & Technology (WEST)
- Women in TPD & Induced Proximity
- AI for All

Publications, patents & talks

Peer-reviewed publications

1. Chemoproteomics discovery of a CNS-penetrant covalent inhibitor of PIKfyve.

   Burton, A. J.*; Chupak, L. S.; Davis, A. J.; Mady, A. S. A.; Meniconi, M.; Teobald, B.; Dorsey, B. W.; Byrne, L. R.; Mulhern, R.; Lundeen, B.; Sorensen, E. W.; Patel, B.; Brennan, S.; Kormocha, D.; Tommasi, R.; Simpson, G. L.; Keillor, J. W.; D'Agostino, L.; Huang, P. S.; Penebre, E.*

   ACS Chem. Biol. **2026**, published online November 6, 2026. doi:10.1021/acschembio.6c00795

2. [2+2] Photocycloadditions of conformationally constrained styrenes enabled by triplet lifetime extension.

   Cong, C.; Fastuca, N. J.; Castanedo, A.; Meinig, P. R.; Woodger, B. C.; Hu, L.; van der Mei, F. W.; Murray, P. R. D.; Davies, G. H. M.; Zoglio, J.; D'Agostino, L. A.; et al.

   J. Org. Chem. **2025**, 90(23), 7751. doi:10.1021/acs.joc.5c00616

3. Identification of small molecule inhibitors and ligand directed degraders of CaMKK1/2.

   Chen, Y.; Whitefield, B.; Nevius, E.; Hill, M.; DelRosario, J.; Sinitsyna, N.; Shanmugasundaram, V.; Mukherjee, D.; Shi, L.; Mayne, C. G.; Rousseau, A.-M.; Bernard, S. M.; Buenviaje, J.; Khambatta, G.; El Samin, M.; Wallace, M.; Nie, Z.; Sivakumar, P.; Hamann, L. G.; McDonnell, D. P.; D'Agostino, L. A.*

   J. Med. Chem. **2023**, 66(23), 15750. doi:10.1021/acs.jmedchem.3c01137

4. Increased CaMKK2 expression maintains fitness of tumor-infiltrating NK cells.

   Juras, P. K.; Racioppi, L.; Mukherjee, D.; Artham, S.; Gao, X.; D'Agostino, L. A.; Chang, C.-Y.; McDonnell, D. P.*

   Cancer Immunol. Res. **2023**, 11(1), 109. doi:10.1158/2326-6066.CIR-22-0391

5. Small molecule allosteric inhibitors of RORγt block Th17-dependent inflammation in vivo.

   Saenz, S. A.*; Local, A.; Carr, T.; Shakya, A.; Koul, S.; Hu, H.; Chourb, L.; Stedman, J.; Malley, J.; D'Agostino, L. A.; et al.

   PLOS One **2021**, 16(11), e0248034. doi:10.1371/journal.pone.0248034

6. Intermolecular crossed [2+2] cycloaddition for polysubstituted 2-oxaspiro[3.3]heptanes.

   Murray, P.; Bussink, W.; Davies, G.; van der Mei, F.; Antropow, A.; Edwards, J.; D'Agostino, L. A.; Ellis, J.; Hamann, L.; Romanov-Michailidis, F.*; Knowles, R.*

   J. Am. Chem. Soc. **2021**, 143(10), 4055. doi:10.1021/jacs.1c01173

7. Discovery of selective, covalent FGFR4 inhibitors with antitumor activity in models of hepatocellular carcinoma.

   Liu, H.; Niu, D.; Tjin Tham Sjin, R.; Dubrovskiy, A.; Zhu, Z.; McDonald, J. J.; Fahnoe, K.; Wang, Z.; Munson, M.; Scholte, A.; Barrague, M.; Fitzgerald, M.; Liu, J.; Kothe, M.; Sun, F.; Murtie, J.; Ge, J.; Rocnik, J.; Harvey, D.; Ospina, B.; Perron, K.; Zheng, G.; Shehu, E.; D'Agostino, L. A.*

   ACS Med. Chem. Lett. **2020**, 11(10), 1899. doi:10.1021/acsmedchemlett.9b00601

8. Evolution of cereblon-mediated protein degradation as a therapeutic modality.

   Chamberlain, P. P.; D'Agostino, L. A.; Ellis, J. M.; Hansen, J. D.; Matyskiela, M. E.; McDonald, J. J.; Riggs, J. R.; Hamann, L. G.*

   ACS Med. Chem. Lett. **2019**, 10(12), 1592. doi:10.1021/acsmedchemlett.9b00425

9. A chiral Ag-based catalyst for enantioselective additions of enolsilanes to α-ketoesters.
   Akullian, L. C.; Snapper, M. L.*; Hoveyda, A. H.*
   J. Am. Chem. Soc. **2006**, 128(20), 6532. doi:10.1021/ja061166o

10. Asymmetric synthesis of acyclic amines through Zr- and Hf-catalyzed enantioselective addition.
    Akullian, L. C.; Porter, J. R.; Traverse, J. F.; Snapper, M. L.; Hoveyda, A. H.
    Adv. Synth. Catal. **2005**, 347, 417. doi:10.1002/adsc.200404319

11. Three-component enantioselective synthesis of propargylamines through Zr-catalyzed additions.
    Akullian, L. C.; Snapper, M. L.*; Hoveyda, A. H.*
    Angew. Chem. Int. Ed. **2003**, 42, 4244. doi:10.1002/anie.200352081

Patents & patent applications

1. Use of bridged cycle-based inhibitors of DNA-dependent protein kinase in combination with DNA polymerase θ inhibitor for gene editing.
   Juno Therapeutics, Inc. WO 2025/160395 A1, July 31, 2025

2. 5,6-Core silacycle based inhibitors of DNA-dependent protein kinase for gene editing.
   Juno Therapeutics, Inc. WO 2025/049247 A1, March 6, 2025

3. Preparation of bridged cycle-based inhibitors of DNA-dependent protein kinase for gene editing.
   Juno Therapeutics, Inc. WO 2025/049253 A1, March 6, 2025

4. Substituted 3-(isoindolin-2-yl)piperidine-2,6-diones for selective degradation of engineered proteins.
   Celgene/Bristol-Myers Squibb. WO 2024/167999, August 15, 2024

5. Substituted imidazo-based compounds as ligand directed degraders of IRAK3.
   Celgene. WO 2024/137743, June 27, 2024

6. 1,2,4-Thiazolidin-3-one-1,1-dioxide inhibitors of protein tyrosine phosphatase.
   Bristol-Myers Squibb. WO 2024/137727, June 27, 2024

7. 1,2,5-Thiadiazolidin-3-one 1,1-dioxide derivatives as PTPN2 inhibitors.
   Bristol-Myers Squibb. WO 2024/137724, June 27, 2024

8. Indazole substituted 1,2,5-thiadiazolidin-3-one derivatives as PTP inhibitors.
   Bristol-Myers Squibb. WO 2024/102791, May 16, 2024

9. 1,2,5-Thiadiazolidin-3-one derivatives as PTP inhibitors.
   Bristol-Myers Squibb. WO 2024/102794, May 16, 2024

10. Substituted imidazopyrazine compounds as IRAK3 binders.
    Celgene. WO 2024/026260, February 1, 2024

11. Substituted imidazopyrazine compounds as ligand directed degraders of IRAK3.
    Celgene. WO 2024/026256, February 1, 2024

12. Substituted pyrazolyl-pyridinyl compounds as ligand directed degraders of IRAK3.
    Celgene. WO 2024/026262, February 1, 2024

13. 5-(2-Fluoro-6-hydroxyphenyl)-1,1-dioxo-1,2,5-thiadiazolidin-3-one derivatives as PTP inhibitors.
    Bristol-Myers Squibb. WO 2023/147531, August 8, 2023

14. Modulators of CaMKK2 as ligand directed degraders.
    Celgene. WO 2023/059873, April 13, 2023

15. Tricyclic fused pyrimidine compounds as HER2 inhibitors.
    Celgene. WO 2023/015240, February 9, 2023
16. Composition and methods for selective degradation of engineered proteins.
    Celgene. WO 2023/015283, February 9, 2023
17. Heterocyclic derivatives as CAMKK2 inhibitors.
    Bristol-Myers Squibb. WO 2022/240826, November 17, 2022
18. Biomolecule conjugates.
    Celgene. US 10,335,495, July 2, 2019
19. Heteroaryl compounds and uses thereof.
    Celgene Avilomics Research. US 9,321,786, April 26, 2016
20. Biocompatible, biodegradable fumagillin analog conjugates.
    US 8,399,512, March 19, 2013
21. Pharmaceutical formulations for fumagillin derivative-PHF conjugates.
    Mersana. WO 2013/096901, December 21, 2012
22. Mutant-selective EGFR inhibitors and uses thereof.
    Celgene Avilomics Research. WO 2012/064706, May 18, 2012
23. Polyal drug conjugates comprising variable rate-releasing linkers.
    Mersana. US 8,524,214, May 27, 2010
24. Biocompatible biodegradable fumagillin analog conjugates for cancer treatment.
    Mersana. WO 2009/073445, June 9, 2009

Presentations & invited talks

1. Introduction to Drug Modalities: an Ever-expanding Toolkit, 2026 Medicinal Chemistry Gordon Research Seminar, lecturer.
2. Induced Proximity from the Med Chem Perspective, 2026 Induced Proximity Gordon Research Conference, session moderator.
3. Designing the 'Holy-Grail' warhead to optimize selectivity, reactivity, affinity & clearance to facilitate development of efficacious, safe covalent drugs, 2025 Covalent Drug Discovery & Development Summit, workshop leader.
4. A Medicinal Chemist's Guide to Drug Modalities, 2025 Medicinal Chemistry Gordon Research Seminar, lecturer.
5. Computational modeling of heterobifunctional degraders of IRAK3, 2024 ACS National Meeting, oral presentation, Olivia Pierce.
6. CaMKK2 inhibitors and degraders, 2022 ACS National Meeting, Mid-Career Investigator Award Symposium, oral presentation.
7. New Approaches in Immuno-oncology, 2022 Medicinal Chemistry Gordon Research Conference, session moderator.
8. New Drugs on the Horizon, 2022 AACR National Meeting, session moderator.
9. New Drugs on the Horizon, 2021 AACR National Meeting, session moderator.
10. Identification of selective orally available small molecule inhibitors and heterobifunctional degraders of CaMKK2, 2021 Medicinal Chemistry Gordon Research Conference, poster presentation.

11. Photosensitized intermolecular [2+2] cycloaddition towards functionalized 2-oxaspiro[3.3]heptanes. 2021, Fall ACS National Meeting, oral presentation by Philip Murray.
12. CNS-penetrant, irreversible covalent HER2 inhibitors for tissue-agnostic treatment of HER2-driven malignancies, 2019 Medicinal Chemistry Gordon Research Conference, poster presentation.
13. Anti-angiogenic and anti-tumor activity of XMT-1107, a fumagillin-derived polymer conjugate, 2009 AACR Annual Meeting, poster presentation.
14. Anti-angiogenic fumagillin-related polymeric prodrugs exhibit antitumor activity in B16 murine melanoma models, 2008 AACR Annual Meeting, poster presentation.
15. Silver fluoride catalyzed enantioselective Mukaiyama aldol additions to ketones, 2006 ACS National Meeting, Division of Organic Chemistry, oral presentation.
16. Catalytic asymmetric synthesis of aryl, alkyl, and alkynyl amines, 2002 ACS National Meeting, oral presentation by John Traverse.